



PROFESSIONAL SUMMARY

Results-driven **Staff AI & Software Engineer** with **9+ years of experience** engineering data-intensive backend platforms, integrating machine learning inference engines, and architecting cloud-native distributed systems at global scale. Extensive experience at Visa building **real-time ML fraud detection systems** processing millions of transactions at 1600+ TPS with strict sub-250ms latency SLAs. Deep hands-on expertise in **Python, Go, Java, and Apache Kafka**, specializing in the MLOps lifecycle from high-throughput streaming data ingestion (Spark/Kafka) to real-time model inference and decision routing. Proven track record of developing GenAI-powered tooling to automate performance analysis and introducing AI-assisted development across global engineering teams. Holds advanced AI/ML certifications from **Stanford University, Caltech, and UT Austin**.

EDUCATION & AI CREDENTIALS

Post Graduate Program in AI & Machine Learning: Business Applications	Dec 2025 – Aug 2026
McCombs School of Business, UT Austin	
AI & Machine Learning Bootcamp	Dec 2024 – Nov 2025
California Institute of Technology (Caltech)	
Machine Learning Specialization	Feb 2023
Stanford University (Online)	
Master of Science, Computer Science	May 2019
George Mason University Fairfax, VA	
Bachelor of Technology, Computer Science	May 2017
Mahatma Gandhi Institute of Technology Hyderabad, India	

TECHNICAL SKILLS

AI & Machine Learning: ML Inference Integration (AIP REST), GenAI tooling (Claude, Cursor, Cline, MCP Servers), Behavioural Analytics (BEAST), Model Retraining Pipelines (EMR), NLP/Prompt Engineering	Cloud (AWS): EC2 (Nitro Enclaves), EKS (Kubernetes), Lambda, EMR, MSK, ElastiCache, S3, API Gateway, KMS
Programming Languages: Python, Go, Java 17, Scala, JavaScript, TypeScript	Databases: Valkey, Redis, Cassandra, Neo4j, PostgreSQL, MySQL, MongoDB
Data Engineering & Streaming: Apache Kafka (MSK), Apache Spark (EMR), Kinesis Firehose, MirrorMaker2, HDFS (Delta Lake)	Infrastructure & DevOps: Kubernetes (EKS), Helm, Terraform / ENV0 (IaC), Jenkins, CodeBuild, Docker
	Security & Architecture: Active-Active Architecture, mTLS, HMAC, AES-GCM, OPA/Rego Rule Engines, Vault

PROFESSIONAL EXPERIENCE

Staff Software Engineer — AI & Cloud Platforms	November 2020 – Present
Visa Inc. Austin, TX	
Engineered full-stack cloud-native services, automated high-throughput data pipelines, and integrated real-time ML fraud scoring engines across AWS and on-premise infrastructure.	
Cloud-Native Fraud Scoring Platform — AWS (Global Payments)	Aug 2024 – Present
Led cloud engineering and ML integration for Visa's dual-region Active-Active real-time payment fraud scoring platform processing UK, Brazil, Argentina, and Costa Rica payment corridors on AWS.	
ML Inference & Decision Service Engineering	
Engineered the VADP Decision Service (Go, EKS, gRPC): executed per-corridor DAG execution integrating AIP ML inference engines , Featurespace ScamDetect models, and OPA/Rego rule evaluation	
Orchestrated real-time Redis GFL/ALR lookups and deduplication TTL behavior (48h Redis TTL) to feed high-velocity transactional data directly into ML scoring models under strict <250ms SLAs	
Architected the Cloud Data Ingestion REST API (Authored Component Operations Map), replacing legacy SFTP batch processing with real-time Kafka streaming to eliminate 24-hour data lag for ML model ingestion	
Data Engineering Pipelines (Kafka / Spark)	
Engineered five Kafka-based data ingestion pipelines (TransactionStatus, Fraud, AccountDetails, Entitlements, HistoricalData) covering JSON streaming, PII encryption (AES-GCM), and real-time data normalization	
Validated ElastiCache → Valkey migration for high-speed in-memory data retrieval feeding the decision engine; confirmed zero latency impact using Valkey Glide client library	
AI Tooling & Team Leadership	
Introduced Claude Code (AI coding assistant) to the A2A Champions team (Feb 2026); configured API access and engineered MCP server integrations (Jira, Wiki, Jenkins) to establish GenAI-driven workflows	
Developed the GenAI Latency Insight Engine : an AI-powered tool analyzing test and execution telemetry to automatically generate actionable performance optimization recommendations	

On-Premise Real-Time Payment Risk Platform

Apr 2023 – Aug 2024

- Validated **43 Risk Indicators (rskInd)** dynamically injected into the ML scoring API response across two phases: mock mode (Consul flag) and live **AIP ML model integration**
- Engineered schema propagation for risk indicators through RaaSOrchestrator, Hadoop consumers (HDFS/Delta Lake), billing, and reporting pipelines
- Led the **AIP Scoring Consumer scalability redesign**: executed Kafka migration (AIP → VRM), upgraded Apache Spark 3.4→3.5, and implemented 4-processor architecture for massive parallel data processing
- Investigated and resolved critical **AIP transaction count mismatches**: traced complex distributed discrepancies across VDP, RaaS HDFS, and the AIP engine, identifying timing and retry-logic root causes

Enterprise Fraud Management — Decision Manager

Nov 2020 – Apr 2023

- Validated **Motif Detectors** integration within the Decision Manager payload; ensured AIP behavioral fraud signals correctly flowed into the decision engine's inputs
- Core contributor to a **Visa-granted USPTO patent**: developed an automated CI/CD pipeline converting Jira ticket state transitions directly into JSON deployment configs, reducing deployment time by 20%

Software Development Engineer

July 2019 – January 2021

FINRA | Rockville, MD

- Led technical validation for **BEAST (Behaviour Analytics)**, CDIP, and CATDD financial surveillance platforms, integrating complex data evaluation logic for regulatory monitoring
 - Automated execution and diagnostics of large-scale financial data processing using Python/Java and custom Linux shell scripts for Jenkins CI/CD pipelines
- Stack:** Java, Python, PostgreSQL, Jenkins, REST APIs, Automation Frameworks, Behavioural Analytics

Full Stack Developer

August 2016 – July 2017

ADP | Hyderabad, India

- Developed real-time chat functionality for IPM HR services using ReactJS and Node.js
 - Optimized web application assets, reducing page load times to under 2 seconds across 10+ websites and boosting conversion rates by 37%
- Stack:** ReactJS, Node.js, MongoDB, RESTful APIs, Webpack

PATENTS & KEY INNOVATIONS

Release Management Automation

USPTO — Granted | Visa Inc.

Co-inventor of an automated pipeline from Jira state transition → JSON config → Jenkins deployment. Reduced deployment time by 20% and eliminated manual steps.

GenAI Latency Insight Engine

Visa Inc.

Created an AI tool analyzing test and execution telemetry to automatically generate actionable performance optimization recommendations for the cloud-native fraud platform.